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Key Points:

- Turbulent heat fluxes are treated as the result of a heat engine operating between the surface and the atmosphere
- Turbulent heat fluxes are determined by maximizing the mechanical power output as an additional thermodynamic constraint
- Despite discrepancies over tropical ocean, broad patterns of turbulent fluxes and temperatures of ERA-Interim can be recovered

Supporting Information:

- Supporting Information S1
- Figure S1
- Figure S2

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Broad climatological variation of surface energy balance partitioning across land and ocean predicted from the maximum power limit

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Abstract Longwave radiation and turbulent heat fluxes are the mechanisms by which the Earth's surface transfers heat into the atmosphere, thus affecting the surface temperature. However, the energy partitioning between the radiative and turbulent components is poorly constrained by energy and mass balances alone. We use a simple energy balance model with the thermodynamic limit of maximum power as an additional constraint to determine this partitioning. Despite discrepancies over tropical oceans, we find that the broad variation of heat fluxes and surface temperatures in the ERA-Interim reanalyzed observations can be recovered from this approach. The estimates depend considerably on the formulation of longwave radiative transfer, and a spatially uniform offset is related to the assumed cold temperature sink at which the heat engine operates. Our results suggest that the steady state surface energy partitioning may reflect the maximum power constraint.

1. Introduction

The Earth's surface is heated by the absorbed solar shortwave and downwelling longwave radiation. To reach steady state, the surface transfers heat into the atmosphere via the emission of radiation and the turbulent fluxes of sensible and latent heat. The relative strength of the turbulent heat transfer from the surface is an important determinant of surface temperature since models in purely radiative equilibrium have significantly higher surface temperatures than models in radiative-convective equilibrium [*Manabe and Strickler*, 1964].

Depending on the spatial and temporal scale of interest, descriptions of turbulent heat fluxes at the surface span a wide range of complexity from the simplest radiative-convective equilibrium models [*Manabe and Strickler*, 1964] to global circulation models (GCMs) and large-eddy simulations [*Smagorinsky*, 1963; *Deardorff*, 1970]. These approaches are commonly computationally expensive or employ semiempirical bulk parameterizations [*Stensrud*, 2009] to determine the turbulent fluxes of sensible and latent heat.

Here we determine the turbulent heat fluxes and their role in shaping surface temperature by asking how their magnitude is thermodynamically constrained. We view the surface-atmosphere system as a heat engine that extracts power from the turbulent fluxes and infer their magnitude from the maximum capacity for sustained mechanical work generation (i.e., power) that maintains atmospheric mixing. There exists an upper limit to the efficiency of heat engines that is termed the Carnot efficiency. For engines operating at this limiting efficiency, the expression for power is proportional to the product of the heat flux and the difference in temperatures at which the heat is added to and removed from the engine. Applied to the surface-atmosphere system, we use the sum of the sensible and latent heat flux as the total heat flux (J) and the difference of the surface (T_s) and atmospheric (T_a) temperatures ($\Delta T = T_s - T_a$). However, J is tightly linked to the energy balances that shape ΔT, resulting in a fundamental trade-off in which a greater heat flux results in a reduced temperature difference. This trade-off results in the maximum power limit proposed before in *Kleidon and Renner* [2013a, 2013b] and *Kleidon et al.* [2015]. With $J = 0$, the temperature difference ΔT is greatest, yielding no power. Yet with a certain value $J = J_{\max}$, ΔT vanishes thus also yielding no power. This implies that power has a maximum value for optimal values, $0 \le J_{opt} \le J_{\max}$ and $\Delta T_{opt} = \Delta T_{opt}(J_{opt})$. This trade-off is critical to our approach and has also been previously used in the context of the proposed principle of maximum entropy production (MEP) [*Paltridge*, 1975; *Ozawa and Ohmura*, 1997; *Lorenz*, 2002; *Ozawa*, 2003; *Kleidon and Lorenz*, 2005].

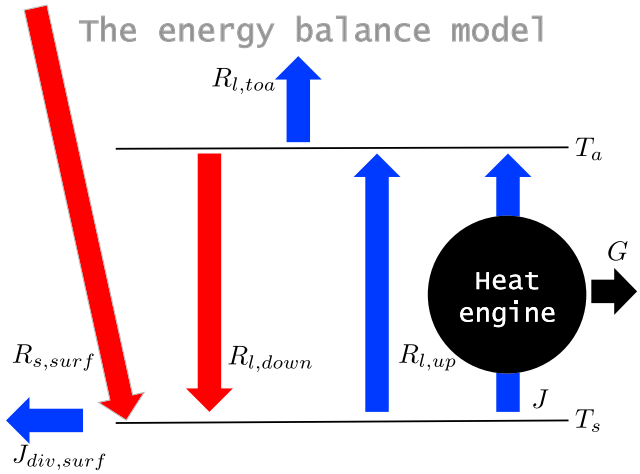

Figure 1. The two-box energy balance model with the atmospheric heat engine generating power to sustain atmospheric mixing. The absorbed shortwave radiation $R_{s,\mathrm{surf}}$, downwelling longwave radiation $R_{l,\mathrm{down}}$, the top of the atmosphere longwave flux to space $R_{l,\mathrm{toa}}$, and the horizontal divergence of oceanic heat transport $J_{\mathrm{div,surf}}$ comprise the forcings, while the quantities we determine are surface turbulent heat flux J, the upwelling longwave radiation $R_{l,\mathrm{up}}$, and surface temperature $T_s = (R_{l,\mathrm{up}}/\sigma)^{1/4}$. The power G is maximized subject to the energy balance constraints. Red colors are used to represent the surface heating fluxes and blue for all effectively cooling fluxes.

We use this limit in conjunction with a simple energy balance model and ask whether the broad climatological patterns and approximate magnitude of the total turbulent heat flux and surface temperature may be estimated. We compare these estimates to an ERA-Interim climatology [*Dee et al.*, 2011]. In doing so, our goal is not to get the most accurate prediction of these fluxes but rather a climatological estimate of the surface energy balance that is based on physical first principles.

2. Methods and Data Sources

We use a simple two-box energy balance model representing the surface and the atmosphere (Figure 1) following *Kleidon and Renner* [2013a]. The turbulent exchange between the two boxes is treated as a heat engine operating at the Carnot efficiency. The expression for power G can be derived from the combination of steady state energy and entropy balances [*Kleidon and Renner*, 2013a] and is given by

$$G = J\left(1 - \frac{T_a}{T_s}\right), \tag{1}$$

where T_s and T_a are the temperatures of the surface and the atmosphere and $\eta_C = 1 - T_a/T_s$ is the Carnot efficiency. In the following, we use the surface energy balance to express T_s as a function of J and show that this can be combined with equation (1) to predict T_s and J in a simple, self-consistent way using the total surface radiative heating as the forcing to our model.

The surface energy budget is described by,

$$R_{\mathrm{in}} = R_{l,\mathrm{up}} + J \tag{2}$$

with $R_{\mathrm{in}} = R_{s,\mathrm{surf}} + R_{l,\mathrm{down}} - J_{\mathrm{div,surf}}$ denoting the total surface heating where $R_{s,\mathrm{surf}}$ is the net solar shortwave radiation at the surface, $R_{l,\mathrm{down}}$ is the downwelling longwave radiation, and $R_{l,\mathrm{up}}$ is the longwave radiation emitted from the surface. J is the total turbulent flux from the surface consisting of the sensible and latent heat fluxes, and $J_{\mathrm{div,surf}}$ is the horizontal divergence of oceanic heat transport. We relate surface temperature T_s to the upwelling longwave radiation by $T_s = (R_{l,\mathrm{up}}/\sigma)^{1/4}$ where $\sigma = 5.67 \times 10^{-8}$ W/m^{-2} K^{-4} is the Stefan-Boltzmann constant. Then the left-hand side of equation (2) involves mostly the total surface radiative heating and the right-hand side the upwelling fluxes from the surface. We use the maximum power limit to determine the partitioning of the fluxes on the right-hand side of equation (2) for a specified forcing consisting of the fluxes on the left-hand side.

Using equation (2) and the Stefan-Boltzmann law for $R_{l,\mathrm{up}}$,

$$T_s(J) = \left(\frac{R_{\mathrm{in}} - J}{\sigma}\right)^{1/4} \tag{3}$$

which combined with equation (1) results in,

$$G(J) = J\left(1 - T_a\left(\frac{R_{\mathrm{in}} - J}{\sigma}\right)^{-1/4}\right). \tag{4}$$

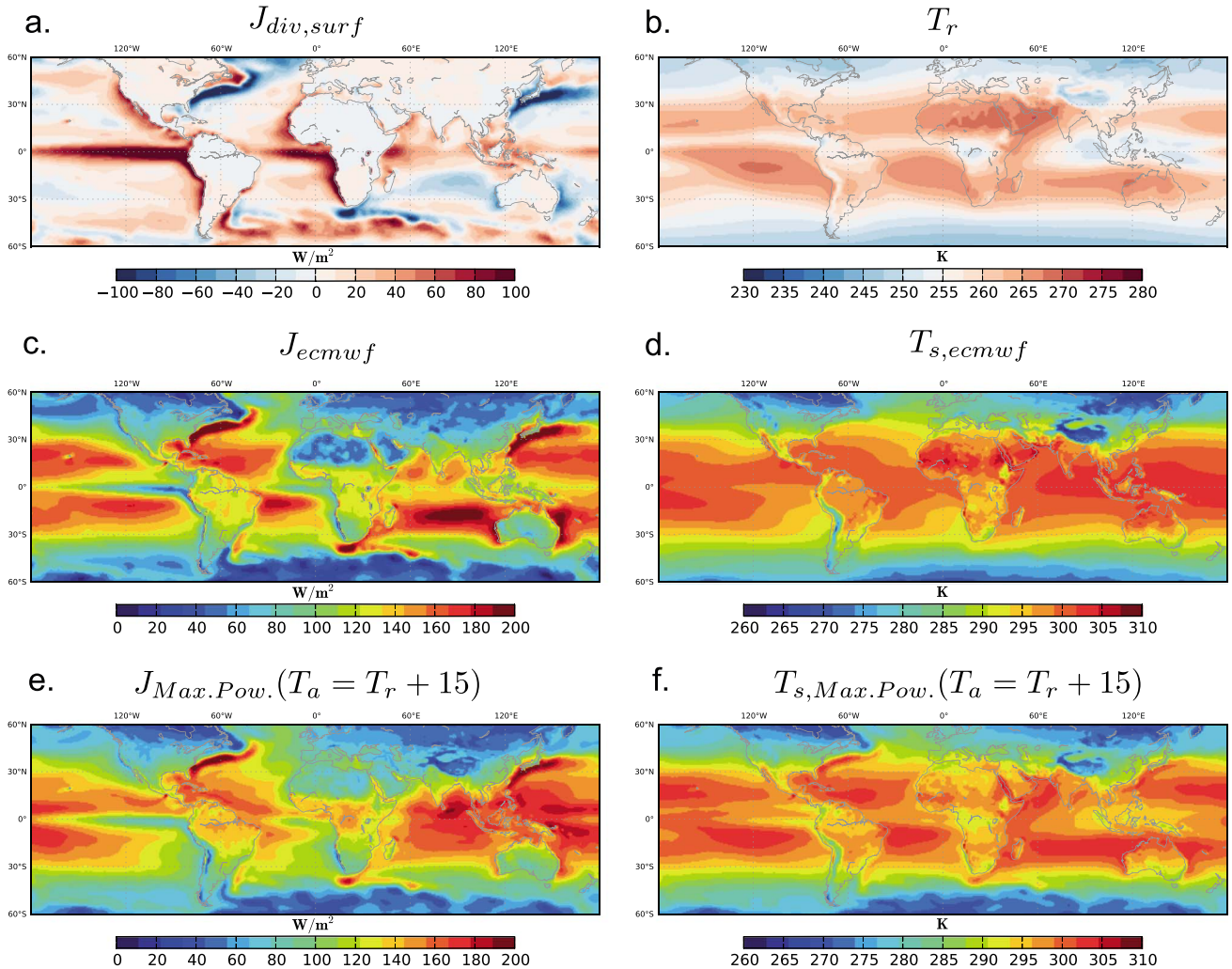

Figure 2. The maps shown are (a) the horizontal divergence of oceanic heat transport $J_{\mathrm{div,surf}}$, (b) the radiative temperature T_r derived from $R_{l,\mathrm{toa}}$, the (c) surface turbulent heat flux and (d) temperature from ERA-Interim, and (e and f) estimates from the maximum power approach having been bias corrected using the atmospheric temperature $T_a = T_r + 15$ K following the discussion in section 3.

The atmospheric temperature appearing in equation (4) is prescribed as the radiative temperature setting $T_a = T_r = (R_{l,\mathrm{toa}}/\sigma)^{1/4}$ where $R_{l,\mathrm{toa}}$ is the total longwave emission to space.

The idea central to our approach is that there exists a maximum value of power for a specified R_{in}. It is easily seen from equation (4) that $G(J = 0) = 0 = G(J = R_{\mathrm{in}} - R_{l,\mathrm{toa}})$, so there is a maximum $G_{\mathrm{max}} = G(J_{\mathrm{MaxPow}})$ for an intermediate value $0 < J_{\mathrm{MaxPow}} < R_{\mathrm{in}} - R_{l,\mathrm{toa}}$. This maximum in G predicts the total turbulent flux J_{MaxPow} at the surface as well as the surface temperature by $T_{s,\mathrm{MaxPow}} = \left(\frac{R_{\mathrm{in}} - J_{\mathrm{MaxPow}}}{\sigma}\right)^{1/4}$.

Due to the nonlinearity of equation (4), we maximize G numerically. T_r is treated as a constant in the maximization since its geographic variation is much smaller than of the surface temperature (see Figure 2). Yet one of the strengths of simple idealized models is that they often allow analytic solutions [*Lorenz and McKay*, 2003; *Robinson and Catling*, 2012]. We find that relaxing the expression of power in equation (4) to a linear function before maximization results in a simple closed form solution, $J_{\mathrm{anly}} = R_{\mathrm{in}}\left[\frac{1}{T_a}\left(\frac{3}{2}\right)^{5/4}\left(\frac{R_{\mathrm{in}}}{2\sigma}\right)^{1/4} - \frac{11}{8}\right]$. The derivation and the solution can be found in Text S3 in the supporting information and prove useful for analytic sensitivity studies such as in Text S4 in the supporting information.

We use the ECMWF Re-Analysis ERA-Interim product [*Dee et al.*, 2011] to determine our forcing R_{in} in order to compute J_{MaxPow} and $T_{s,\mathrm{MaxPow}}$. We then evaluate how well our approach predicts the patterns of turbulent fluxes and surface temperatures by comparing these to the relevant fields in the ERA-Interim product. Since

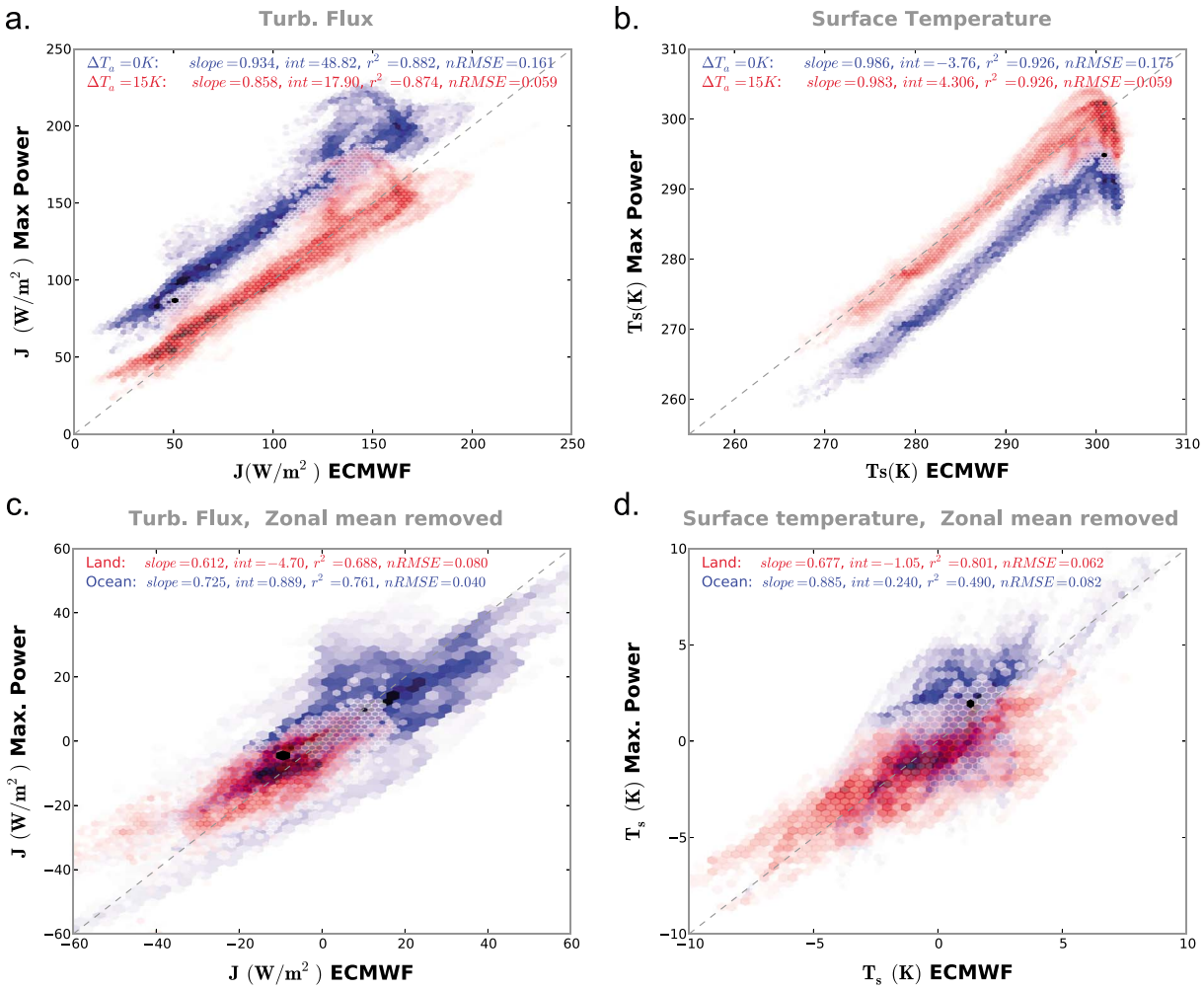

Figure 3. Comparison of the estimates from maximum power to ERA-Interim. (a and b) Turbulent heat flux and surface temperature, respectively. Points in blue result from using $T_a = T_r + \Delta T_a$ where $T_r = (R_{l,\mathrm{toa}/\sigma})^{1/4}$ and $\Delta T_a = 0$ K and the ones in red from using $\Delta T_a = 15$ K. (c and d) Similar to Figures 3a and 3b but with zonal means removed. Land (red) and ocean (blue) are separated. Only $\Delta T_a = 15$ K is shown here since $T_a = T_r$ overlap almost exactly with these. Darker colors in all plots represent higher density of grid points.

$J_{\mathrm{div,surf}}$ is not a standard output of ERA-Interim, we approximate it from the local imbalance in the surface energy budget, $J_{\mathrm{div,surf}} = R_{s,\mathrm{surf}} - ((R_{l,\mathrm{up}} - R_{l,\mathrm{down}}) - J_{\mathrm{ecmwf}})$ (Figure 2). Note that while $J_{\mathrm{div,surf}}$ includes the surface upwelling fluxes, it contains no information on their partitioning. These data are available at a resolution of 0.75° × 0.75°, and climatological means were computed from long-term annual means of monthly values for the period 1979–2014.

Further details on the methodology are provided in Text S1 in the supporting information.

3. Results and Discussion

Maps of the geographic variation of J_{ecmwf} and $T_{s,\mathrm{ecmwf}}$ as well as J_{MaxPow} and $T_{s,\mathrm{MaxPow}}$ are shown in Figure 2. It can be seen that our approach explains the broad patterns seen in the ECMWF output despite significant deviations in certain regions of the tropical Indian and the western Pacific oceans. These maps reveal that the largest magnitudes of J are found along the warm Gulf Stream and the Kuroshio Current in both our estimates and ERA-Interim while the lowest magnitudes are along the cold Humboldt and Benguela currents. We quantitatively study their correlation in Figure 3, limiting this study to the latitudinal range to 60°N to 60°S since the heat engine approach does not account for the possibility of temperature inversions at the higher latitudes.

Table 1. Importance of the Parameterization of Longwave Fluxes to the Broad Patterns of the Turbulent Heat Flux Inferred From Maximizing Power[a]

#	Parameterization of $R_{l,\mathrm{down}}$	Solution		Slope	Intercept	r^2	nRMSE (%)
1	$R_{l,\mathrm{down}} = \mathrm{data}$	Numeric	Global	0.93	48.8	0.88	16.1
			Land	1.06	37.77	0.85	25.3
			Ocean	0.91	51.82	0.88	16
2	$R_{l,\mathrm{down}} = \mathrm{data}$	$J = R_{\mathrm{in}}\left(\frac{1}{T_a}\left(\frac{3}{2}\right)^{5/4}\left(\frac{R_{\mathrm{in}}}{2\sigma}\right)^{1/4} - \frac{11}{8}\right)$	Global	1.08	40.2	0.87	19.2
			Land	1.21	28.1	0.86	27.6
			Ocean	1.05	44	0.86	19.6
3	$R_{l,\mathrm{net}} = k_r(T_s - T_a)$	$J = R_{s,\mathrm{surf}}/2$	Global	0.51	27.72	0.86	12.8
			Land	0.41	43.52	0.5	14.2
			Ocean	0.57	18.7	0.98	13.8

[a]The statistics derive from the comparison of J_{MaxPow} to J_{ecmwf} (Figure 3) for the different scenarios enumerated. The first scenario (row 1) comprises the comparison of J_{MaxPow} with J_{ecmwf} using $R_{l,\mathrm{down}}$ specified by ERA-Interim data. The second scenario (row 2) compares the given analytic solution with J_{ecmwf}. The last scenario (row 3) compares results of the simplified two-box model of *Kleidon and Renner* [2013a] that uses a linear approximation for the net longwave radiation $R_{l,\mathrm{net}} = k_r(T_s - T_a)$, where k_r may be interpreted as a linearized radiative conductance. The first two rows clearly capture geographic variation significantly better, and the magnitude of the correlation r^2 is more consistent for both land and ocean comparisons. The root-mean-square errors have been normalized by the range of the respective data. Note that biases in T_s are largely equivalent and opposite to J. Note that the data are limited to 60°N–60°S.

The blue points in Figures 3a and 3b show the comparison of the maximum power limit to the ERA-Interim product. The estimated turbulent heat flux as well as surface temperature correlate strongly with ERA-Interim ($r^2 = 0.88$ and 0.93 for J and T_s) and show slopes close to one (= 0.93 and 0.99) signifying that our formulation estimates very well the broad variations of both. The analysis is extended in Table 1 (scenario #1) to demonstrate the applicability of this approach over both land and ocean.

However, we find that our results are systematically biased, which reflects in the nonzero intercepts (48.8 W/m^2 and −3.76 K) in the respective comparisons. We note that since J and T_s are related through equation (3), bias in one results in an equivalent opposite bias in the other as can be seen in Figure 3. We attribute this bias to our assumption that the heat engine operates with the radiative temperature inferred from $R_{l,\mathrm{toa}}$ as the cold temperature of the heat engine. It is shown analytically in Text S4 in the supporting information that the analytic expression J_{anly} can be used together with the observed intercept (48.8 W/m^2) to directly compute the offset minimizing the bias and is found to be $(\Delta T_a)_{\mathrm{min}} \approx 13.6$ K. We perform a numeric sensitivity analysis modifying the atmospheric temperature to $T_a = T_r + \Delta T_a$ for $\Delta T_a = 0, 5, 10, 15$, and 20 K. The results of this study (Table S2 in the supporting information) confirm the offset inferred from the analytic solution showing that the bias indeed changes uniformly with ΔT_a and is at a minimum for $\Delta T_a = 15$ K. The plots in red in Figure 3 include this bias correction, for which the root-mean-square error (RMSE) value is reduced by ≈65% for both J and T_s. In summary, the turbulent fluxes and surface temperatures of the ERA can be seen as the result of the maximum power output of an ideal heat engine at the Carnot efficiency with an effective atmospheric temperature $T_a \approx T_r + 15$ K. There are different interpretations possible for why the temperature offset is required:

1. *Overestimation of the effective height of convection*. The region of atmospheric mixing where work is performed is proportional to $T_s - T_a$ and can be related to the height of convection. Choosing $T_a = T_r$ appears to overestimate the surface-atmosphere temperature contrast that drives the heat engine and thus the effective height of convection [*Manabe and Strickler*, 1964] and the magnitude of J.
2. *The nonblack nature of the Earth's atmosphere*. Since the real atmosphere has a vertical gradient of opacity, the atmospheric temperature may be expressed more appropriately as $T_a = (R_{l,\mathrm{toa}}/\epsilon_a\sigma)^{1/4}$ where $\epsilon_a < 1$ is the effective emissivity of the atmosphere. Thus far, we had implicity set $\epsilon_a = 1$. For the choice of $\epsilon_a \approx 0.8$ one obtains $T_a = T_r + 15$. Thus, the temperature offset may be related to the atmosphere not being a black body.
3. *The real efficiency of the atmospheric heat engine being lower than the Carnot efficiency due to internal dissipation*. This is discussed further in Text S5 in the supporting information. In the context of the simple

approach we have outlined in this work, one cannot resolve which of these may be the most appropriate interpretation.

We further evaluate how well our model explains the flux and temperature patterns in the zonal direction to show that the high correlation is not simply due to the strong zonal variation in solar radiation. The plots of Figures 3c and 3d evaluate the correlation between the residuals removing zonal means. The slopes and correlations decrease yet remain significant with $r^2 = 0.8$ and $r^2 = 0.64$ for J and T_s showing that the high correlation is not due to the large zonal variations alone.

Application of the model at grid scale uses the horizontal divergence of oceanic heat transport as an extra forcing (Figure 2) as it cannot be diagnosed within our model. An explicit computation of atmospheric radiative transfer, clouds, greenhouse, and aerosol forcing is not performed here; thus, $R_{l,\mathrm{down}}$ must be specified as forcing.

We also evaluate the role of the downwelling longwave radiation ($R_{l,\mathrm{down}}$) to our results. The results discussed thus far involve specifying $R_{l,\mathrm{down}}$ from reanalysis data and the numeric maximization of power. In this formulation, the atmospheric box represents the integrated effects of the atmosphere on the downwelling longwave radiation at the surface. This is the most physically representative case within the idealized structure of our model. Rows 1 and 2 of Table 1 list statistics of the comparison of the numeric (J_{MaxPow}) and analytic (J_{anly}) solutions with J_{ecmwf}. For global means, the slope of the correlation changes by only ≈15% suggesting that the analytic solution reveals the approximate functional dependence of the turbulent heat flux on the forcing, making it useful for analytic sensitivity studies.

However, earlier work [*Kleidon and Renner*, 2013a] formulated the maximum power approach with an alternative parameterization. Linearizing the Stefan-Boltzmann law around a reference temperature T_{ref}, it was shown that the optimal solution to maximizing power is the simple expression, $J = R_{s,\mathrm{surf}}/2$, the approach requiring fewer parameters. In this formulation, the atmospheric box represents not the integrated effect but an unspecified, single level in the atmosphere and cools by emitting uniformly in the upward and downward directions. Comparisons analoguous to those discussed above are found in row 3 of Table 1 and show that the reduction in slope is considerable being ≈45% despite high correlations. These approximate solutions were usefully employed to study the surface heat fluxes over land [*Kleidon et al.*, 2014], and the sensitivity of the hydrologic cycle [*Kleidon and Renner*, 2013b], however, poorly captures the flux and temperature patterns analyzed in this work.

We attribute the significantly greater success of the direct use of $R_{l,\mathrm{down}}$ to it incorporating the effects of atmospheric radiative transfer and surface forcing induced by cloud, water vapor, and greenhouse gases [*Goody*, 1964; *Philipona et al.*, 2004]. In other words, the forgoing discussion demonstrates the importance of the magnitude and parameterization of the longwave fluxes to the outcome of the maximum power limit.

Our approach relates closely to the previously proposed principle of maximum entropy production (MEP) [*Paltridge*, 1975; *Ozawa*, 2003; *Kleidon and Lorenz*, 2005] which states that nonequilibrium thermodynamic systems relax to a state that maximizes the rate of entropy production. MEP has previously been used to explain mantle convection [*Vanyo and Paltridge*, 1981; *Lorenz*, 2002], meridional heat transport [*Paltridge*, 1975; *Lorenz et al.*, 2001], and globally averaged atmospheric temperature and convective flux profiles in a 1-D model [*Ozawa and Ohmura*, 1997]. Maximizing power is equivalent to maximizing entropy production in the two-box model when the power is assumed to be fully dissipated at the cold temperature. However, the advantage we see of the maximum power formulation is a more intuitive interpretation of the physical limits prescribed by the second law of thermodynamics. Moreover, the magnitude of the predicted power could be tested, for instance, by comparing the power to the generation term of the TKE budget of the convective boundary layer [*Moeng and Sullivan*, 1994].

The highly idealized formulation outlined here is clearly subject to several limitations. The key assumptions we make are the following: (A) steady state and (B) that the heat engine operates at the Carnot efficiency. We now discuss the validity of each of these. We approximate steady states using long-term annual means (1979–2014). This averages out most seasonal and interannual climate variability while heat storage is accounted for by the $J_{\mathrm{div,surf}}$ term. Yet variations with longer time scales such as the global warming signal may persist. This may introduce a bias in our comparisons but are expected to be small relative to the magnitudes of the surface fluxes.

Assumption (B) is a strong assumption although it has been used in previous studies [*Rennó and Ingersoll*, 1996; *Emanuel and Bister*, 1996], since internal dissipation within the heat engine is expected to result in losses in efficiency of power. The potential limitations of an idealized heat engine approach have been discussed in later works such as *Pauluis and Held* [2002b, 2002a], while *Laliberteé et al.* [2015] discuss a general methodology to compute entropy budgets from GCM output and estimate inefficiencies in atmospheric power. Yet our argument fundamentally rests on the coupling between the turbulent flux and surface temperature defined by the surface energy balance rather than the efficiency that has been assumed. Indeed, we expect that our results are to hold also for lower efficiencies that are similar or proportional to the Carnot efficiency since these are expected to result in a similar J-T_s trade-off. We demonstrate quantitatively in Text S5 in the supporting information that our results are robust to a significant relaxation of the assumption of the Carnot efficiency.

Despite these limitations, we have demonstrated that our minimalist approach captures the broad variation of the surface energy partitioning across land and ocean based on the total surface heating.

4. Summary and Conclusions

We have formulated a simple two-box energy balance model constrained by maximum power to predict the broad variation and approximate magnitudes of the turbulent heat fluxes and surface temperatures in the climatological mean. These magnitudes were determined from the maximum power predicted partitioning between the radiative and turbulent heat fluxes at the surface for a prescribed solar radiative forcing, downwelling longwave radiation, and the horizontal divergence of oceanic heat transport. We also quantified the importance of the parameterization of downwelling longwave radiation to our results.

Despite several limitations, our estimates of turbulent fluxes and surface temperatures correlate very well with ERA-Interim reanalysis output. Our idealized heat engine approach is distinguished from more complex formulations by the absence of empirical parameterizations. The surface energy partitioning is instead determined by the hypothesis that the system operates near the thermodynamic maximum power limit. Our results suggest that the turbulent heat fluxes and surface temperatures may indeed be broadly determined by this limit in the climatological mean thus providing useful first-order baseline estimates.

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